

Customer Credit Default Prediction

AAI500 - Applied Artificial Intelligence

San Diego University

Team Members:

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Instructor: Azka Azka

Team Members & Their Contributions

Anugrah Rastogi

- Data Preparation
- EDA
- Plot and Analysis

Dhrub Satyam

- Designed and tuned all ML models
- Performed GridSearchCV and interpreted classification metrics

Mallesham D

- Final analysis
- Conclusions and Report

Abstract

- **Aim:**

- Predict customer credit default using machine learning.

- **Data:**

- 70K training, 33K test records with 80+ features.

- **Methods:**

- Data cleaning, One hot encoding, RobustScaler.

- **Models:**

- Logistic Regression, Decision Tree, Random Forest, XGBoost.

- **Outcome:**

- Random Forest performed best in accuracy and interpretability.

- **Suggestion:**

- Deploy Random Forest with periodic fairness and performance audits.

Introduction

Problem

Credit default prediction helps financial institutions manage risk.

Objective

Build ML models using behavioral & demographic data to predict defaults.

Dataset

Public Kaggle dataset (2023) with various personal and financial attributes.

Goal

Develop interpretable, accurate, and deployable predictive models.



Real-World Motivation

- Banks need to detect defaulters before credit approval.
- Reduces financial losses, speeds up approvals, and improves compliance.
- We used machine learning to automate and improve this prediction process using real customer data.

Data Handling & Preparation

- **Dataset Overview**

- 70,000 training records with 82 features
- 33,000 test records with 80 features.

- **Missing Values**

- Filled missing numeric values using median

- **Categorical Features**

- One hot encoded

- **Feature Scaling**

- Used RobustScaler to minimize effect of outliers.

- **Target**

- 0 - Non-default, 1 - Default.

- **Feature Alignment**

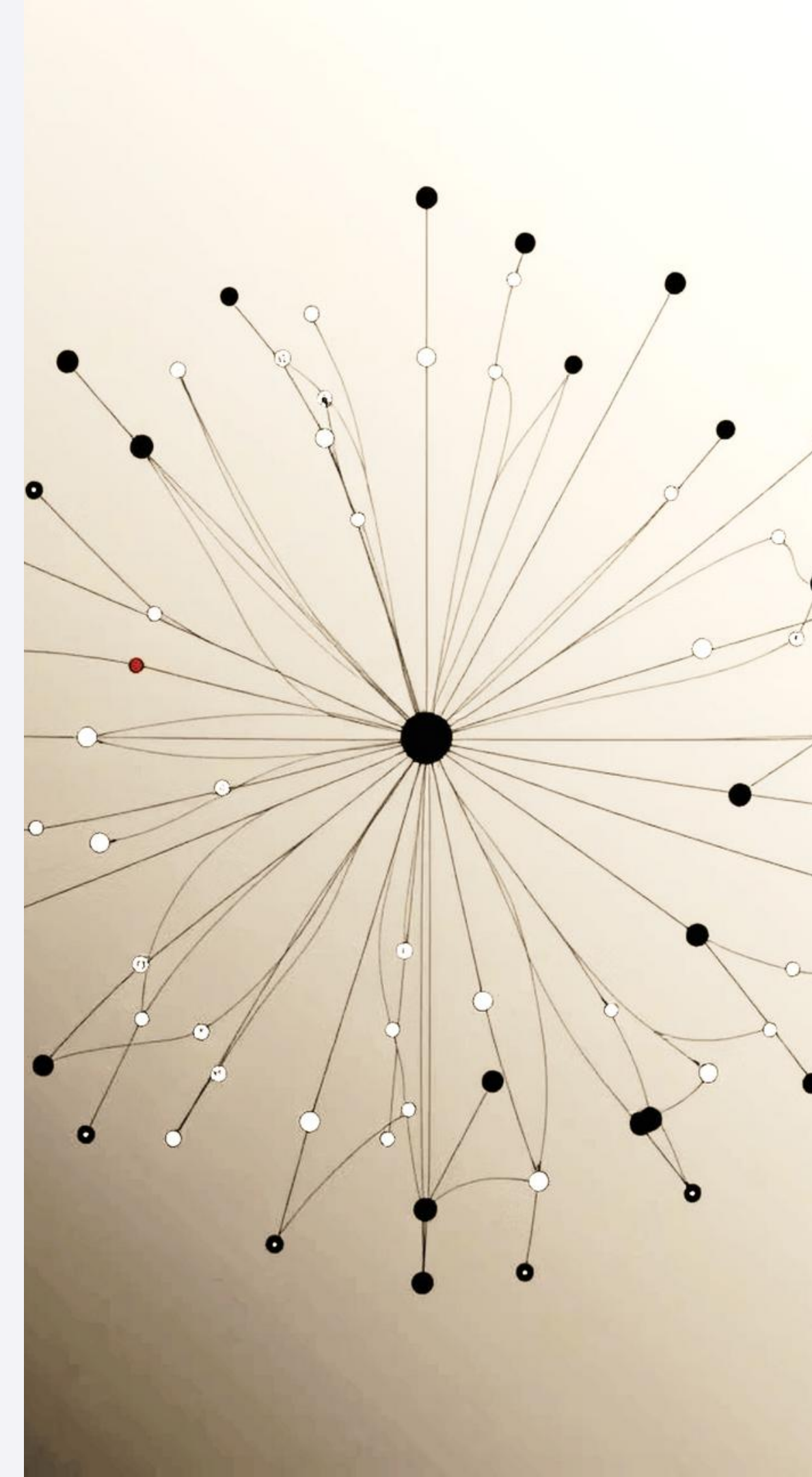
- Ensured consistency across training and test datasets.

Exploratory Data Analysis

- **Univariate:** Skewed income & debt ratios; high outliers.
- **Bivariate:** Default correlates with high debt_ratio and dependents.
- **Target Distribution:** Imbalanced (Default: 20.6%, Non-default: 79.4%).

Actions Taken:

- Used class weights to handle imbalance.
- Applied robust scaling.
- Chose tree-based models due to low correlation among features.
- Used boxplots, heatmaps, and PR curves for insights



Feature Selection

- **Why?**
 - Original dataset contained 189 features
 - Reduces overfitting and model complexity
 - Improves training speed and generalization
 - Helps eliminate noise and irrelevant information
- **Technique 1: Information Value (IV)**
 - Measures strength of a feature in separating target classes
 - 85 features removed due to low or suspicious IV values
- **Technique 2: Pearson Correlation**
 - Identifies multicollinearity between features
 - Highly correlated features ($\geq 75\%$) are redundant
 - 16 features removed due to high pairwise correlation

Model Selection Overview

- **Models Tested:**

- Logistic Regression (baseline, interpretable)
- Decision Tree (visual, simple)
- Random Forest (ensemble, robust)
- XGBoost (high accuracy, regularized)

- **Tuning Tool:**

- GridSearchCV with 5-fold CV, optimized for ROC AUC.

- **Metrics:**

- ROC AUC, F1, Precision, Recall

Classification Metrics



Precision

- Of predicted defaulters, how many were correct?



Recall

- How many actual defaulters were identified?



F1 Score

- Balance between Precision & Recall.



ROC AUC

- Ability to distinguish between default & non-default across thresholds.

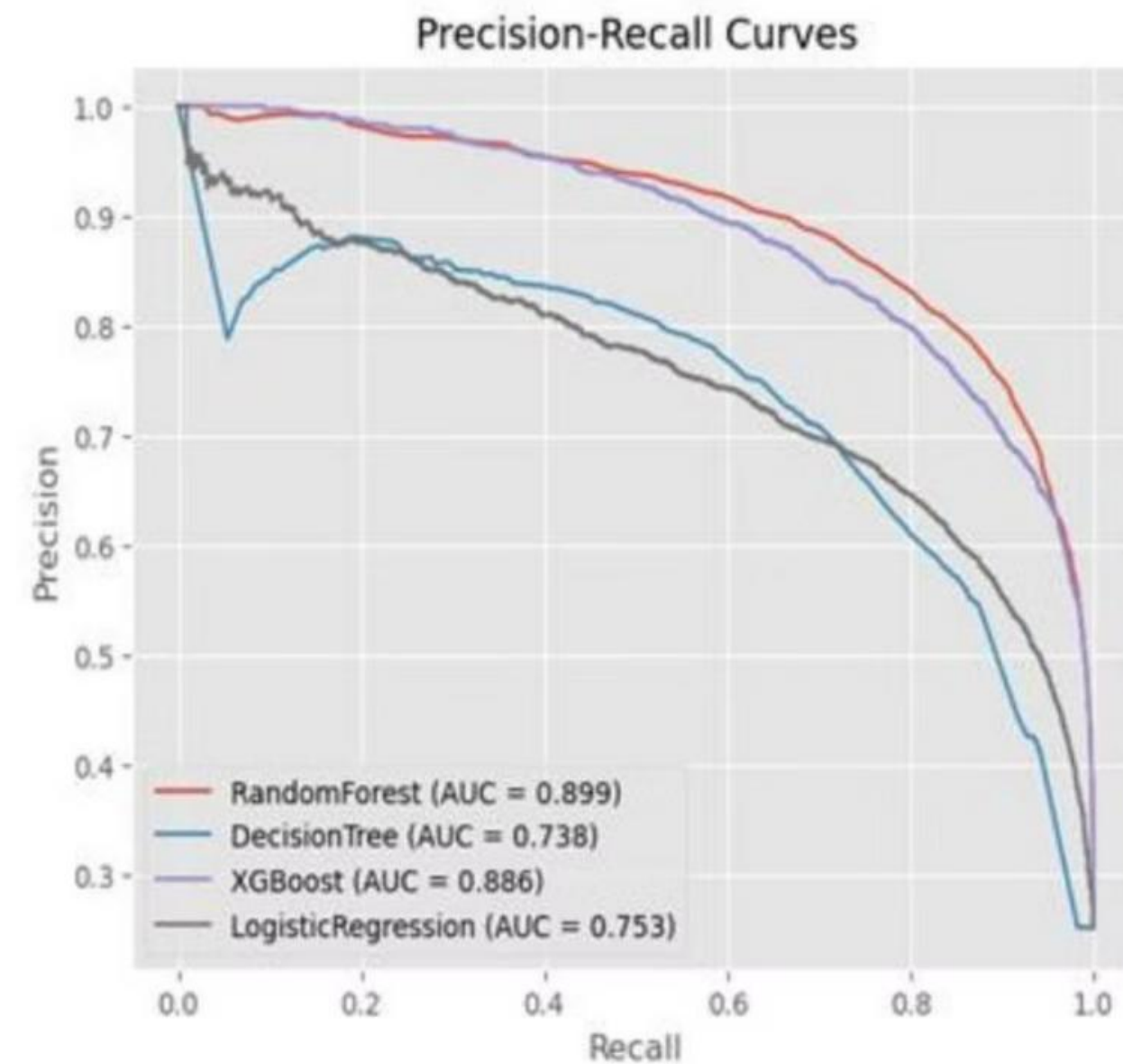
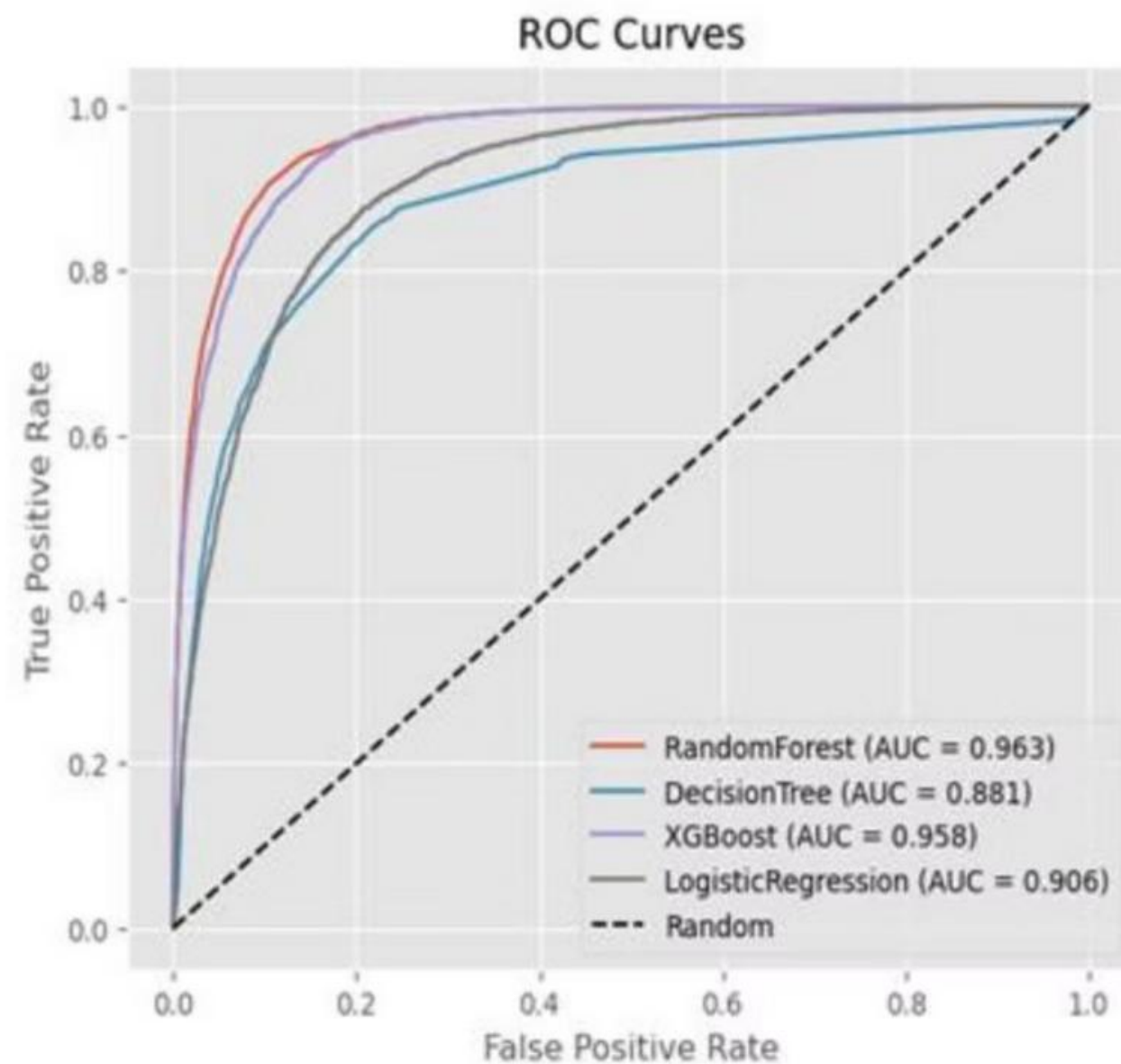
Model Performance (Validation Results)

Model	Accuracy	ROC AUC	F1 Score (Class 1)	Precision (Class 1)	Recall (Class 1)
Random Forest	0.91	0.9633	0.8049	0.85	0.76
XGBoost	0.90	0.9576	0.7906	0.82	0.76
Decision Tree	0.85	0.8808	0.6855	0.75	0.63
Logistic Regression	0.83	0.9057	0.7134	0.63	0.82

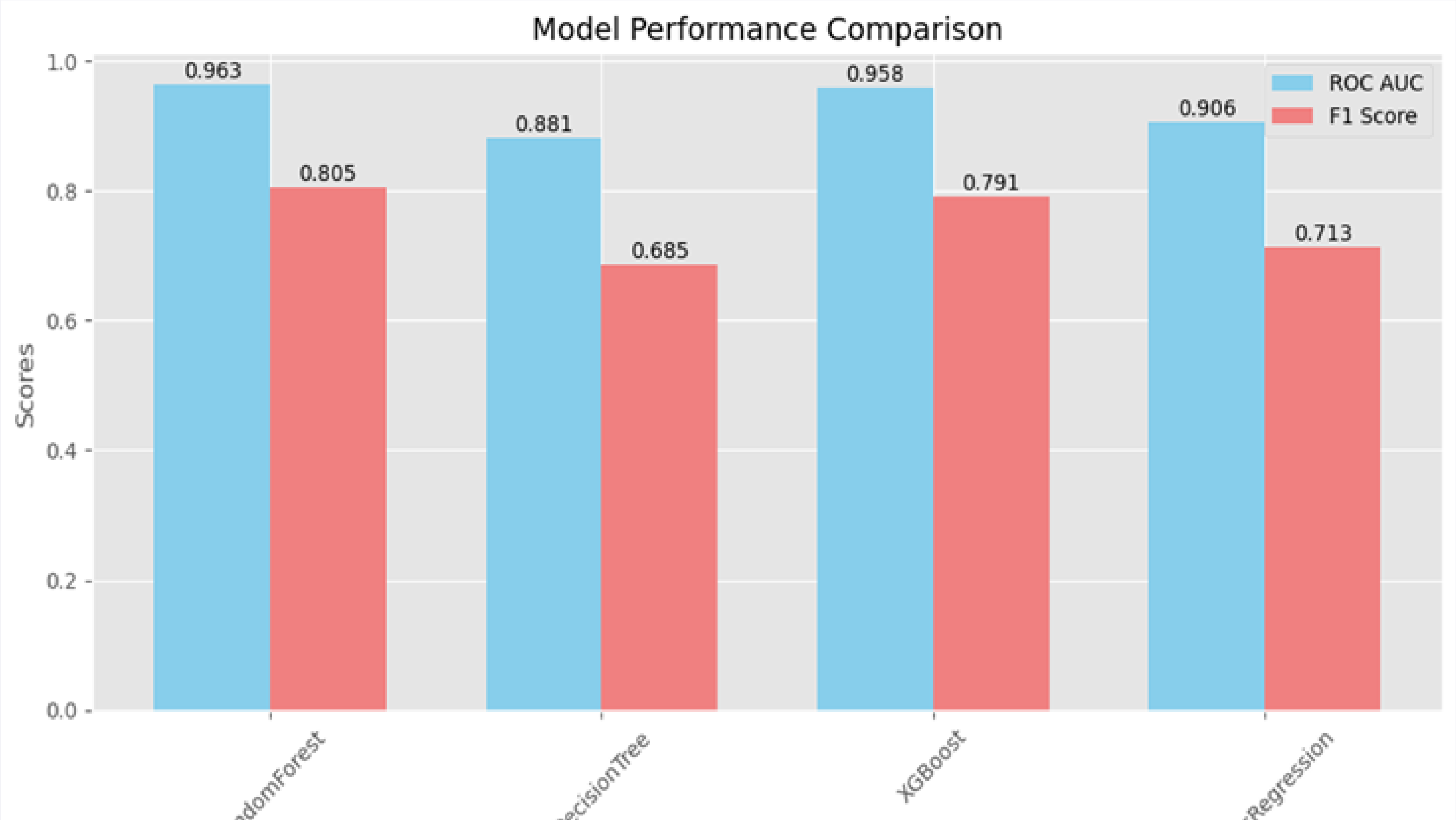
Best model: **Random Forest** due to high ROC AUC and F1 score.

Visualizations

- **ROC Curve:** Illustrates the trade-off between True Positive Rate and False Positive Rate.
- **Precision-Recall Curve:** Useful for imbalanced datasets, focusing on the minority class.
- **Model Comparison Bar Chart:** Visually compares ROC AUC and F1 scores across all models.



Visual Results





Final Model - Random Forest

- Most reliable in detecting defaulters and avoiding false positives.
- Handles noise and outliers effectively.
- Suitable for mixed data types (numerical + categorical).

Best performance on:

- Accuracy: 91%
- ROC AUC: 0.9633
- F1 Score: 0.8049

Business Impact

- **Benefits:**
 - Early identification of high-risk customers
 - Faster loan approvals for low-risk profiles
 - Supports compliance with explainable decisions
 - Cuts operational cost via automation.

Final Recommendations

- Deploy Random Forest model in production.
- Implement:
 - Continuous monitoring
 - Fairness and bias audits (e.g., gender, age groups)
 - Periodic tuning
 - Explainability tools like SHAP / LIME
 - Advanced ensemble techniques (stacking RF + XGBoost)

Conclusions

- Built multiple models to predict credit default.
- **Random Forest** emerged best for deployment.
- Logistic Regression: Good recall, poor precision.
- Decision Tree: Too simple.
- XGBoost: Promising alternative, slightly behind Random Forest.

